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DS PROGRAMMING

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Depth First Search

DEPTH FIRST SEARCH TRAVERSAL

Develop a C program to represent a graph using an adjacency matrix and perform Depth First Search (DFS) traversal starting from a given source vertex. Print the sequence of visited vertices and explain how DFS explores the graph.

```
1 #include <stdio.h>
2
3 int main() {
4     int n, s;
5     scanf("%d", &n);
6     int a[n][n], vis[n];
7     for (int i = 0; i < n; i++) {
8         for (int j = 0; j < n; j++) scanf("%d", &a[i][j]);
9         vis[i] = 0;
10    }
11    scanf("%d", &s);
12
13    int st[n], top = -1;
14    st[++top] = s;
15    vis[s] = 1;
16
17    printf("DFS: ");
18    while (top >= 0) {
19        int v = st[top--];
20        printf("%d ", v);
21    }
```

OUTPUT

Your INPUT goes here!

Your INPUT goes here!